Prospective consequential LCA: **the environmental consequences of micro/meso decisions in a world in transition**

BROADER ECONOMY IN TRANSITION · CONTEXT SCENARIOS (SSP/RCP)

Micro- and meso-level decisions in diverse contexts
designing, operating, deploying or adopting technologies, with environmental consequences that unfold over the transition

which units operate?

how they operate?

Pillar I · Market consequences
Which suppliers respond, when and where, with what environmental profile

Pillar II · Process consequences
How processes and their environmental flows respond to design, operation and context

FROM KNOWN TO UNKNOWN

Develop and evaluate against a reference model

Confront with observed and reported data

Extend where no full reference exists

One consistent representation of market responses, from full models to simple heuristics PhD I	**Joint approach** same perturbations of models and success criteria in both pillars	**Processes as interoperable model classes that respond to context, operation and maturity** PhD II
Testing marginal-supplier shortcuts against full model responses PhD I	**Energy** Pillar I case	**AI-assisted model building, evaluated where the answer is known** PhD III
Checking marginal-supplier methods against capacity actually built MSc I	**Metallurgy** shared case	**Testing whether vintage-resolved classes change results from generic database data** MSc II
Disaggregating global scenarios to the local markets where decisions play out MSc III	**Nitrogen** shared case	**Representing an emerging process before deployment, as alternative configurations** PhD III

OUTCOMES

Scenario-consistent marginal suppliers
and the errors of common shortcuts, characterized

Comparison across cases
market and process effects

Interoperable process-model classes
and an evaluated workflow to build more

Environmental decision support across contexts
how much representation choices change the environmental conclusions, at the market and at the process level

MEANS

Integrated assessment and energy-system models, SSP/RCP scenarios, market data	AI, to make systematic modelling and testing affordable	Open process-model infrastructure, engineering models, LCA databases